# LAKSHMINATH K

Chennai, India

Software Developer — Simulation & Immersive Training

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in/lakshminath-k | Lakshminath97/portfolio-site

## PROFILE

Software Developer specializing in XR simulations, immersive training solutions, and interactive 3D applications, backed by a strong foundation in Mechanical Engineering. Proficient in Unity3D, C#, OpenXR, and XR Interaction Toolkit, with experience integrating software, hardware, and IoT systems to deliver end-to-end simulation solutions. Previously automated CAD workflows using NX Open, improving design efficiency by 40%. Proven ability to drive flagship projects from concept to deployment, lead cross-functional development teams, and translate complex operational requirements into scalable, immersive training experiences.

## EXPERIENCE

### Senior Software Developer
2026 – Present

Spatiotech

- Delivered immersive training modules, including a flagship VR forklift simulation featuring a custom real-time simulator rig.
- Led the flagship project from ideation to deployment, integrating Unity/C#, Meta Quest 3, OpenXR, ESP32, REST APIs, and Supabase.
- Lead an 8-member cross-functional team across software development and hardware engineering, overseeing technical execution, coordination, and delivery.
- Developed and integrated LMS capabilities to track trainee progress, training completion, and evaluation performance.
- Deliver end-to-end enterprise XR solutions, from client requirement analysis and SOP structuring to development, integration, and deployment across VR platforms.

### Engineer
2022 – 2026

Valeo India Pvt. Ltd.

- Designed automotive HVAC components and complex assemblies using CATIA V5, applying DFM/DFA principles, GD&T, and injection-moulding design practices.
- Managed CAD data and version control using ENOVIA V6 and performed vibration-jig simulations to identify potential failure points.
- Developed a C#-based CAD automation tool using NX Open to generate 3D screw models, technical drawings, and PDFs from standard screw-size selections, improving design efficiency by 40%.
- Developed Unity3D-based XR factory simulations to support immersive design reviews and visualize manufacturing environments.

### Engineer
2021 – 2022

SACHA Engineering Pvt. Ltd.

- Delivered production-ready BIW fixtures, grippers, and welding tools for global automotive programs, supporting manufacturing readiness and production release.
- Performed clamp load capacity and cantilever behaviour analyses to validate fixture designs and identify potential structural issues before production.
- Supported immersive design reviews with global engineering teams by preparing VR-ready 3D models for virtual design validation.

### Graduate Engineering Trainee
2019 – 2021

SACHA Engineering Pvt. Ltd.

- Designed BIW fixtures using CATIA V5 and Siemens NX 12 for automotive manufacturing programs.
- Contributed to an XR project deployed at Ford in 2019, modelling factory components from point-cloud data and supporting development of a VR factory walkthrough using Autodesk Inventor, Navisworks, and Unity3D for immersive factory visualization and design reviews.

## SKILLS

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|---|---|
| **Programming** | C#, Python, Git, MVC architecture, SOLID / DRY principles, REST API and Supabase integration |
| **XR & Real-time 3D** | Unity3D, OpenXR, XR Interaction Toolkit, Meta All-in-One SDK, SteamVR, AR Foundation |
| **Hardware & IoT** | Meta Quest 3, Pico, HTC VIVE, PC VR, ESP32, simulator-rig integration, on-site deployment |
| **CAD & PLM** | CATIA V5, Siemens NX, NX Open API, Autodesk Inventor, Navisworks, ENOVIA V6 |
| **Professional** | Client requirement analysis, SOP structuring, team leadership, technical documentation, AI-assisted development |

## EDUCATION

**B.E. Mechanical Engineering** — Velammal Engineering College · CGPA 8.81 — 2015 – 2019

## AWARDS & RECOGNITION

- **Best-in-Class VR Training** — Client Recognition
- **Customer Centric Award** — Valeo (Dec 2024)
- **Best Team Player** ×**3** — Valeo (2024–25)
- **Consistently Delivers Excellence** ×**2** — Valeo (2024–25)

## CERTIFICATIONS

- C# Unity 3D Game Development in Unity 6 (Udemy)
- NX Open API Training with C# .NET (Feesworth)
- CAD Automation with Excel (Udemy)
- Master Diploma in Product Design (CADD School)